Query-First Semantification of CEUR-WS Proceedings Metadata in Wikidata

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Abstract. CEUR Workshop Proceedings (CEUR-WS) has been a non-profit publisher for scientific workshop and conference proceedings in computer science since 1995. For more than 30 years, it has used HTML and PDF as the single source of truth for the metadata required by indexers such as DBLP and the German national library DNB. Scholars have long requested persistent identifiers for the core entities involved: event series, events, proceedings, papers, editors, authors, institutions, and publishers, with the goal of allowing semantic handling and querying with state-of-the-art tools.
In this work, we show how we enabled the semantification using Wikidata as the target knowledge graph infrastructure. The 2015 Semantic Publishing Challenge queries are finally permanently available with up-to-date metadata. The Scholia portal provides user-friendly and aspect-oriented access and is widely used.
We also point out the challenges of aligning the digital, social, and physical aspects of the ongoing transformation and outline a roadmap to use the available queries to bootstrap a new RDF-based metadata-first publishing approach that will mine metadata from proceedings matter and generate the HTML from it instead of the other way around.

Keywords: Knowledge Graph · Linked Data · Metadata Extraction · Publishing · Semantic Web · Wikidata

1 Introduction

Traditional publishing workflows often handle the metadata of the required core entities (cf. Figure 1[4]) in a manner that does not comply with the Findable, Accessible, Interoperable, and Reusable (FAIR) principles[5], giving low priority to these principles as well as to the Single Source of Truth (SSoT) and Single Point of Truth (SPoT) principles of data management [35]. In this work, SSoT

[4] blue entities are the first milestone for the CEUR-WS Wikidata bot
[5] We refer to such workflows as "unFAIR"; see Section 3.1

Fig. 1. Most relevant entities for scientific publishing (abstract schema).

Fig. 2. SWAT4HCLS2024 example graph walk

refers to the one authoritative source from which metadata originates, whereas SPoT refers to the one authoritative location at which that metadata is accessed by downstream consumers. Therefore, the opportunities to collect the metadata during the lifecycle of an event [20] are missed. As technical editors of CEUR Workshop Proceedings, we have been feeling this pain for over a decade. This work shows how the original HTML/PDF SSoT of CEUR-WS at the ceur-ws.org domain (SPoT) has been used to create a Knowledge Graph representation of the relevant entities in Wikidata between 2022 and 2026 as a step in this direction, thus allowing portals such as Scholia to present the RDF query results in an aspect-oriented and user-friendly way.

1.1 CEUR Workshop Proceedings

Since 1995, CEUR Workshop Proceedings (CEUR-WS)[6] has published more than 4,200 Volumes and over 90,000 papers as a free online service for publishing proceedings of scientific workshops (and smaller conferences) in computer science, operated by a team of unpaid volunteers.

Technically, all data and metadata of the proceedings are directly represented as a static website using HTML, PDF and HTTP and FTP protocols as provided by the workshop editors along the CEUR-WS "how to submit" guide [7].

The metadata is available after some delay of weeks or months through indexing services such as DBLP [27] and K10plus [39][7] and at the archive of the TIB `https://www.tib.eu/`.

We report on the successful use of Wikidata and Scholia as the knowledge graph platform to provide FAIR metadata for semantic analysis and querying.

1.2 Contributions of this Paper

The core contributions presented in this work are as follows:

1. To provide the tools, infrastructure, and approaches to consistently supply high-quality FAIR metadata (semantification) for CEUR-WS as detailed in section 3.
2. Splitting the metadata for the three main entities: Event, Event Series, and Proceedings is a major step towards improving the metadata quality and FAIRness.

[6] `https://ceur-ws.org/`

[7] `https://dblp.org/`, `https://opac.k10plus.de`

3. Providing a bootstrapping approach [3,13] to shift the single point of truth from the current HTML/PDF files to an RDF ready ontology-based format as outlined in Section 6.

These contributions are valuable for other publishing use cases that struggle with the same pain points.

2 From Semantic Publishing Challenge 2014 via Text2KG 2023 to Production 2026

The Semantic Publishing Challenge [26,12,37] started in 2014 with Task 1 calling for "answering queries related to the quality of workshops by computing metrics from data extracted from their proceedings, also considering information about persons and events".

The relevance of the original set of 20 queries was subjectively rated by us, resulting in a list sorting the queries by priority[8]. The most relevant queries and another 5 queries that rely on the main index (Q1.5, Q1.12, Q1.13, Q1.16, and Q1.17) have been taken as the definition of done. We reported on the successful Wikidata export of all CEUR-WS proceeding and event metadata in 2023 at the TEXT2KG Workshop [20]. The SPARQL queries[9] are available as named parameterized queries [19].

3 CEUR-WS Semantification in Wikidata

This section describes how the CEUR-WS metadata is extracted from its HTML/PDF single point of truth into the Wikidata knowledge graph. Section 3.1 motivates the semantification from the perspective of the FAIR principles and derives the requirements on queries, ontologies, curation and infrastructure. Section 3.2 then presents the concrete extraction pipeline built around `pyCEURmake`, Cermine, GROBID and `snapquery`, and the `ceur-spt` prototype that consumes the resulting graph. Figure 3 gives an overview of the overall workflow.

3.1 Making CEUR-WS FAIR

The FAIR principles [24,42] with Persistent Identifiers (PIDs) and rich metadata as core components are being adopted by an increasing number of publishing organizations.

To make CEUR-WS metadata FAIR, a set of relevant queries should be supported, as derived from the original set of queries of the 2014 Semantic Publishing Challenge as outlined in Section 2. The metadata should be provided based on established ontologies. The manual and automatic curation of entries should

[8] https://cr.bitplan.com/index.php/List_of_Queries
[9] https://cr.bitplan.com/index.php/Semantic_Publishing_Challenge_Queries

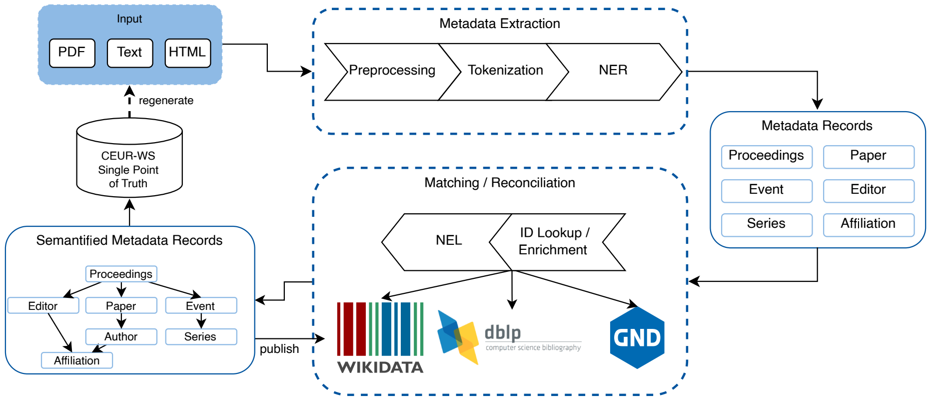

Fig. 3. Workflow of the CEUR-WS Semantification

be possible with public access for all stakeholders, e.g., editors, authors, organizers, publishers, and indexers. The infrastructure should be stable and there should be sufficient trust in its long-term availability. An open source non-profit infrastructure is preferred since this is also the mode of operation of CEUR-WS.

Given the HTML/PDF/text input of CEUR-WS, the corresponding Knowledge Graph needs to be created and the single-point-of-truth computer-readable metadata has to be separated from the different representations such as HTML so that the above requirements are fulfilled.

Both the HTML and PDF encoding of the original scientific content are structured for the purpose of optimizing the display / output on paper or screens; therefore, there is a structure loss compared to what was originally available in the document source files the authors might have been using [14]. Most scholars are not aware of this loss in the daily use of published content just because the documents are optimized for display and human consumption [44,2]. For the metadata extraction and use in knowledge graphs the difference is sometimes disastrous, e.g., when a simple text from a PDF document cannot be extracted any more due to exotic styling and formatting or simply because only a scanned graphic image version of an older document is available that needs optical character recognition to extract the textual content.

The metadata needed for creating the knowledge graph is only available in natural language/text form and follows rules that have been changed multiple times during the history of CEUR-WS. From 2013 to 2026, there have been 33 different versions[10] of the index file template with no proper tracing of what was changed from version to version. The index and volume HTML files were often edited manually, leading to minor undocumented differences even within these versions. The usage of the different versions follows a long-tail Zipfian distribution with 5 versions covering 60% of all volumes.

[10] https://cr.bitplan.com/index.php/CEUR-WS_Versions

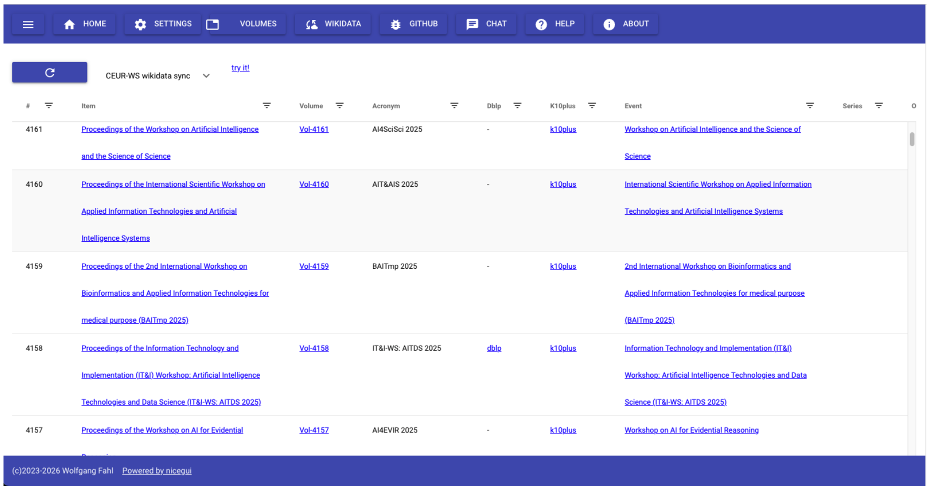

Fig. 4. CEUR Volume Browser wikidata synchronization with downstream dblp and k10plus coverage

3.2 Extracting metadata from HTML and PDF targeting Wikidata

pyCEURmake [17] was created to parse the main HTML index as well as the individual proceedings HTML pages, vastly improving on an earlier simple scraper [29]. The results have been cached in JSON and SQLite storage for quick access. Location entries have been consolidated using the geograpy3 library [36]. Author, Editor and Paper details have been aligned with the DBLP SPARQL endpoint.

Cermine [38] and GROBID [28] have been run on the papers to provide additional metadata references. Figure 4 shows a user-friendly presentation of the data mined by these steps.

For each core entity, we analyzed the availability of Wikidata properties by using our wdgrid analysis tool [16,22], which also generates SPARQL queries. We aligned the queries with the stated needs of the Semantic Publishing Challenge.

snapquery [19] was used to make sure that the queries are compatible with different endpoints and do not only run on the official blazegraph Wikidata query endpoint.

The ceur-spt prototype [18] demonstrates the benefits of semantification by improving linking and navigation capabilities compared to the current static CEUR-WS HTML pages. A complete Cermine run has been integrated so that paper-level metadata (authors, affiliations, references) is available in the per-volume pages as illustrated for Vol-3262 in Figure 7.

4 Challenges in Aligning the Digital, Social, and Physical Aspects

CEUR-WS is a digital-native, online-only publisher operated by volunteer editors, and serves a scholarly community of editors, authors and research institutions. Downstream indexing activities are handled by DBLP, the German National Library (DNB), TIB, Wikidata/WikiCite and the Scholia community. This section describes the challenges of aligning the digital, social, and physical aspects, since traditional proceedings are still printed on paper and stored on book shelves in libraries.

The three-way alignment is necessary to avoid cognitive dissonance in the discussions and decisions that are needed to support the metadata-first roadmap outlined in Section 8.

4.1 Digital: CEUR-WS as a native online publisher

CEUR-WS has always been online only, without physical artefacts. As discussed in Section 3.1, the current Single Point of Truth encodes metadata in HTML and PDF optimized for printing and human display rather than for machine consumption. There has been a good reason to never run any software such as a triplestore—static content is much more stable and safer than infrastructure that needs maintenance. The Wikidata export makes it possible to query metrics such as coverage and consistency. The public synchronization dashboard[11] is an example that makes the downstream indexing observable; see Figure 4.

4.2 Social: volunteer ecosystems across institutional boundaries

CEUR-WS editors, DBLP curators, DNB and TIB librarians, Wikidata and WikiCite contributors, and Scholia maintainers all operate as volunteer or mixed-funding communities, each with their own goals and procedures. Coordination across such communities is loosely coupled and based on the trust that what worked in the past will continue to do so in the future. Our work highlights gaps; for example, DBLP is currently indexing only about 80 % of all volumes. Trust in Wikidata as an openly editable knowledge graph has yet to be established, and active checks are needed to detect whether the metadata has been tampered with in unacceptable ways.

4.3 Physical: library catalogues and the shelf model

Library catalogues such as the DNB-curated integrated authority file GND[12] historically optimize for physical books — proceedings that sit on a shelf. For a digital-native publisher there is a mismatch: the distinction between *event*, *event series* and *proceedings*, which the core-entity schema of this paper treats as

[11] https://cvb.wikidata.dbis.rwth-aachen.de/wikidatasync

[12] https://www.dnb.de/EN/Professionell/Standardisierung/GND/gnd_node.html

three separate first-class entities (Figure 1), is not covered adequately in library catalogues. Our effort to convince the DNB to systematically curate event series based on an existing concept in its ontology has so far been unsuccessful.

5 In-Use Impact – WikiCite, Scholia, CEUR-WS

This section reports on the in-use impact of the CEUR-WS semantification in three overlapping communities. WikiCite is an initiative to create a bibliographic database based on Wikidata [40].

Scholia [31] is a portal serving Wikidata content for the needs of scholars.

WikiCite, Wikidata, Scholia and the CEUR-WS stakeholder community are overlapping communities. There are over 100,000 distinct authors and over 6,500 distinct editors with digital traces in the DBLP CEUR-WS index entries.

5.1 Scholia

It is based on a set of over 380 named parameterized SPARQL queries organized around 40 different aspects. The Wikidata Scholarly Articles Subgraph Analysis [1] showed that roughly half of the 17 billion triples that form the main graph belong to scholarly articles and their linked entities. The Wikimedia foundation needed to perform a graph split due to a 4 Terabyte maximum storage limit of the Blazegraph backend in use. The Scholia community decided to use QLever [4] as the new backend and make all queries SPARQL 1.1 compliant [43]. Note that before the graph split, there was a hold on adding more scholarly articles to Wikidata which was the reason we did not mass-insert CEUR-WS paper metadata yet. Writes to Wikidata are performed through the officially approved bot account *User:CEUR-WS* [8], for which the community permission was obtained in 2023 [41].

Table 1 shows 12 SPARQL-driven panels on the Scholia event aspect [31]. All queries are enabled by this work and rendered identically by the legacy Blazegraph endpoint at `https://scholia.toolforge.org/event/Q124669329` and by the post graph-split QLever endpoint [4,43] at `https://qlever.scholia.wiki/event/Q124669329`, illustrated for SWAT4HCLS 2024 (Q124669329)[13]. All 12 panels survived the 2025 Wikidata graph split unchanged; only the underlying engine and the SPARQL dialect (1.1 compliant) differ.

Figure 5 shows the user-friendly display of the coauthorship as graph visualization.

5.2 WikiCite

WikiCite [40] provides the bibliographic backbone on which CEUR-WS metadata is contributed to Wikidata. Since the CEUR-WS Wikidata bot was activated, the community members contributed to the bot created entries.

[13] https://www.wikidata.org/wiki/Q124669329

Table 1. Scholia SWAT4HCLS 2024 panels based on SPARQL queries

#	Panel/Query	style	entities
1	`data`	header	event
2	`people`	table	scholar
3	`co-authors`	graph	scholar
4	`sponsors`	table	institution
5	`timeline`	timeline	event
6	`topic-scores`	table	
7	`uses`	table	
8	`recent-publications`	table	paper, scholar
9	`proceedings`	table	proceedings, paper
10	`presentations`	table	
11	`related-events-timelocation`	table	event
12	`related-events-people`	table	event,scholar

Figure 6 shows the distribution of Wikidata editors who have created or revised CEUR-WS *proceedings* items (conference proceedings (Q1143604)[14]). The long-tail distribution illustrates that while the officially approved *User:CEUR-WS* bot [8] accounts for the bulk of systematic mass contributions, a broad community of individual WikiCite editors continuously curates, corrects and enriches the proceedings metadata. This overlap of bot-driven bulk ingestion and human community curation is a core strength of the WikiCite approach and a prerequisite for the Scholia panels discussed in Section 5.1 above. The plot was generated with `ceur-wd-contributions` [17] by traversing the revision history of all CEUR-WS proceedings items on Wikidata.

5.3 CEUR-WS stakeholders

The CEUR-WS bot [8] currently exports *proceedings* and *event* items only. Items for the related entities event series, paper, scholar and institution are created and edited by the CEUR-WS stakeholders that are active in the Wikidata community. The SWAT4HCLS 2024 case has already been presented in Section 1 (graph walk, Figure 2) and in this Section 5 (Scholia panels, Table 1 and Figure 5).

6 Single Point of Truth Bootstrapping

The work reported so far uses the CEUR-WS HTML/PDF artefacts as the single point of truth (SPoT) and derives the Wikidata knowledge graph from them. This section outlines how that relationship is being inverted in a bootstrapping fashion [3,13], so that a machine-readable, ontology-based representation becomes the SPoT and the HTML/PDF artefacts are generated from it. Section 6.1 introduces a first draft of the CEUR-WS Proceedings Ontology.

[14] https://www.wikidata.org/wiki/Q1143604

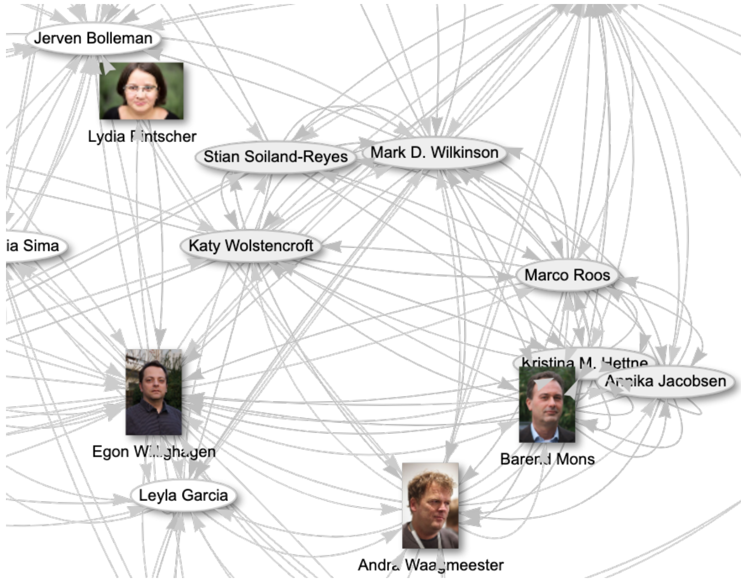

Fig. 5. SWAT4HCLS 2024 co-author graph visualization

6.1 CEUR-WS Proceedings Ontology

The CEUR-WS Proceedings Ontology formalizes the structure of the semantified data. A first draft is available and covers all entity classes defined by the Semantification Challenge (see Figure 8). To maximize reusability and interoperability, terms from established ontologies are reused wherever appropriate, including the SPAR Ontologies [32], Dublin Core, and Schema.org; crosswalks to Wikidata and DBLP are provided. The draft is already operational: JSON-LD representations are generated for every volume.[15] Ongoing schema consolidation is driven through two open discussions in the public `semantify` repository, covering the schema itself [10] and sample volume subgraphs [9].

Four design questions remain open: Four design questions remain open. (i) **Vocabulary selection.** Widely adopted vocabularies such as Schema.org and Dublin Core are preferred over bespoke terms, with mappings provided where necessary; the precise choice is still under discussion for several entity types. (ii) **Model granularity.** Open issues include the representation of session structure and the distinction between joint proceedings and joint events. (iii) **Tooling.** Generation of the JSON-LD from the workshop website and PDFs should be automated, so that editors curate only the JSON-LD and the HTML page is regenerated from it, inverting the current HTML-as-SPoT situation described in Section 3.1. (iv) **Shape-based validation.** SHACL [25] or ShEx [34] shapes should be generated from the ontology to encode editor experience as machine-checkable rules, including cardinalities, permitted value sets, and required iden-

[15] e.g. `http://ceurspt.wikidata.dbis.rwth-aachen.de/Vol-3264.jsonld`

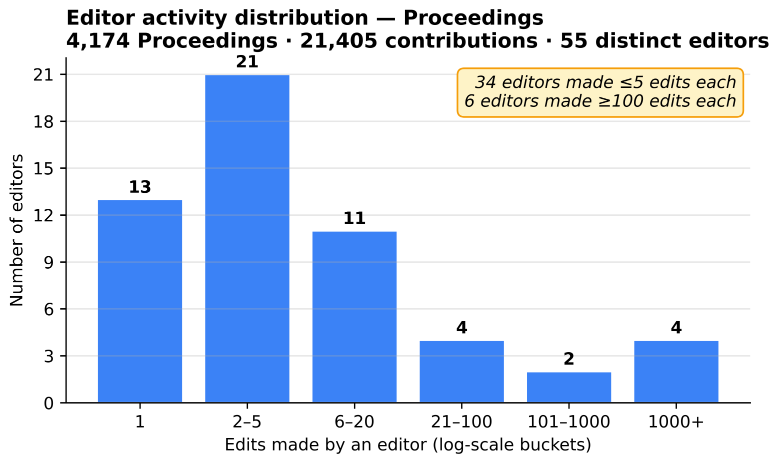

Fig. 6. Editor activity distribution across 4,174 CEUR-WS proceedings items on Wikidata, showing the complementary roles of automated bulk ingestion and targeted human curation of the WikiCite community (55 distinct editors, 21,405 total contributions).

tifiers. Beyond ingestion-time validation, shapes can drive form generation for the editor tooling in (iii), serve as machine-readable documentation for downstream consumers, and enable continuous quality monitoring of the published knowledge graph.

7 Lessons Learned

The effort for this work required was substantially higher than initially anticipated, as a large number of non-trivial edge cases in the legacy metadata had to be resolved individually and manually.

7.1 Queries as first-class citizens

The Semantic Publishing Challenge and the Scholia portal both use queries as the core structuring elements. Using the guiding principle "If you can not query it, it does not exist" changes the view on how to provide metadata. While traditionally schemas and ontologies are created first to allow querying for use-case specific areas of interest, the query-first and query as first-class citizens approach describes the information needs of stakeholders as queries. Having the queries as named, parameterized, technology-independent entities themselves is helpful [19]. Nielsen's Synia [30] approach shows a lightweight application of this idea.

A main success factor for this work was being able to apply metadata to the queries. This way the queries could be ordered by task and/or relevance and

Fig. 7. Vol-3262 ceur-spt page: structured paper-level metadata derived from the HTML index and PDF mining pipeline (cropped)

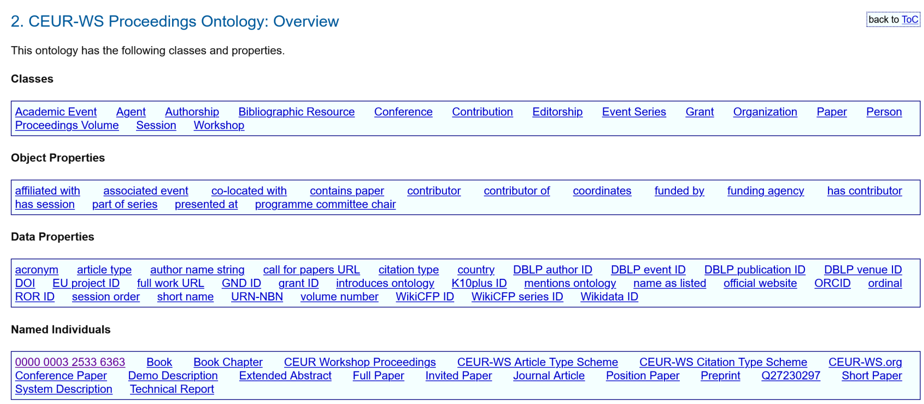

Fig. 8. Overview of the proposed CEUR-WS Proceedings ontology

linked to their respective implementation. These links were just motivations - as first-class citizens the queries are available with proper URIs as persistent identifiers and part of a meta-knowledge graph that is useful for the further semantification steps.

7.2 Property creation as a community process

Wikidata properties such as colocated with (P11633)[16] undergo a Wikidata's property proposal process[17] before a new property is created. As described in [20] the creation of property proposal[18] took some effort and led to the clarification

[16] https://www.wikidata.org/wiki/Property:P11633
[17] https://www.wikidata.org/wiki/Wikidata:Property_proposal
[18] https://www.wikidata.org/wiki/Wikidata:Property_proposal/colocated_with

that the property is asymmetric. The property was considered as highly relevant and well defined.

7.3 Legal aspects of Publishing Personal Data of Scholars

The proposed workflow calls for publishing personal data of authors and editors early in the life cycle of an event. According to the legal advisors we consulted, this is legal since a combination of conditions (Art. 6 GDPR) is met: Explicit consent is given by the stakeholders (Art. 6(1)(a)), the legitimate interest of the stakeholders is pursued (Art. 6(1)(f)), the data might be public already, the processing is necessary for scientific research (Art. 6(1)(e), Art. 89). The strongest condition is that there might be a legal obligation (Art. 6(1)(c)) as would be the case if publishing via major print publishers that have to provide a copy of the proceedings to the German National Library according to German law. Consent, Public Data, and Contractual Necessity are crucial for scholarly publishing. The legal basis and data-subject rights for publication of personal data of scholars by CEUR-WS are documented in the CEUR-WS privacy policy[19] and tracked in the project issue tracker issue #11 of the CEUR semantify project[20].

7.4 Incremental migration over one-shot replacement

The challenge in introducing the new CEUR-WS workflow is to find a sweet spot between effort and benefit [33]. Given the long-tail distribution of ever more exotic corner cases that have shown up in the analysis of the existing data and workflow there is potentially a huge list of "small" problems that would require an inadequately high effort to be solved systematically. The "sweet spot" is in solving only the most relevant problems and not fixing some of the exotic details at all or doing it manually. Therefore we will avoid a "big bang" introduction, which would replace the old approach in one single big (and therefore very risky) step. Instead, we propose to make smaller changes gradually. As a first step, the main index.html shall be split by years (with a link to the old legacy complete list). Then, the per-year list may be generated from the single point of truth data. This will be continued iteratively, going from the current year back to a point where it seems no longer reasonable to spend further effort.

A first bootstrapping prototype of this year-split is already available (see Figure 9).

8 Roadmap: Metadata-First Publishing

The coverage of the core entities Event Series, Papers, Scholars and Institutions needs to be improved by consolidating and completing the RDF data of the downstream indexing services at DBLP and German National Library (DNB).

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[19] `https://ceur-ws.org/privacy-dsgvo.html`

[20] https://github.com/ceurws/semantify/issues/11

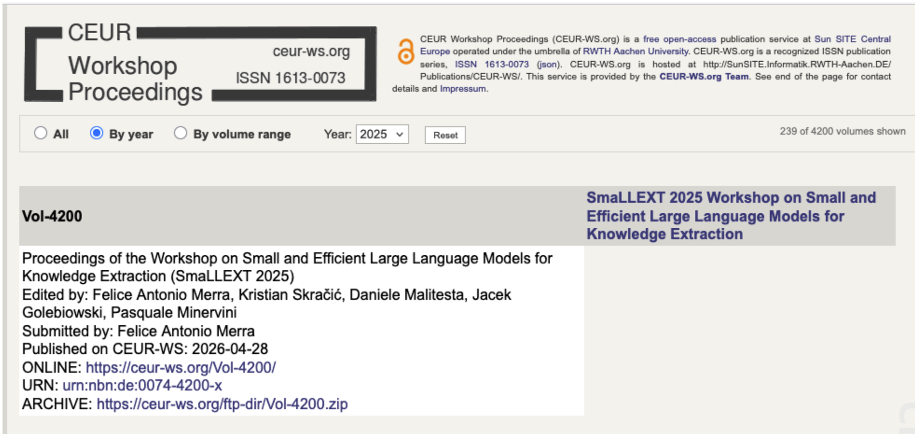

Fig. 9. Year-split prototype of the CEUR-WS index (`index_alt.html`): a step towards iterative replacement of the monolithic index page.

The central long-term objective of this effort is to shift the CEUR-WS single point of truth from the current HTML/PDF artefacts to a workflow in which the metadata is mined from the proceedings matter and the rendered HTML and computer readable RESTful content-negotiation artefacts should be generated from it as static content. The 2026 migration of the CEUR-WS infrastructure from RWTH Aachen to Gesellschaft für Informatik provides the opportunity to take further concrete steps in this direction. The open-source CEUR-WS semantify project [23] coordinates this work.

8.1 Community proposals for the future workflow

Several complementary proposals have been provided during the timeframe of this work:

- *Author-driven metadata entry*: Michael Cochez proposes re-using the `ceur-ws-prototype` [11], inspired by `ceur-make-ui` [6], with a simple HTML form so that authors enter their own metadata, thus automating volume creation and reducing editor workload.
- *Automated extraction and compliance checking*: Jerven Bolleman proposes automating metadata/PDF extraction, index generation and CEUR compliance checks using `ceur-index-html-helper` [5], already applied to generate `https://www.jerven.eu/swat4hcls-proceedings-2026/`.
- *SPT-first research prototype*: the authors propose making the ceur-spt backend fully functional first and only then adding the frontend, following the `SemPubFlow` prototype approach.
- *Historical context*: the 2019 XML-based proposal by Sarven Capadisli and the ESWC 2022 background discussion by Nathanael Arndt are acknowledged as previous ideas and work.

9 Conclusion

CEUR-WS proceedings metadata is systematically made available in Wikidata via the CEUR-WS bot. The relevant Task 1 Named Parameterized Queries of the Semantic Publishing Challenge of 2014/2015 are now available on different endpoints namely the SPARQL 1.1-compatible QLever endpoint. The Scholia portal provides production access to queries based on and extending this set. The Wikidata, WikiCite, Scholia, and CEUR-WS stakeholder communities have appreciated and contributed to this work.

9.1 Limitations

The contributions reported here are subject to several limitations. First, Wikidata coverage is currently complete for the *proceedings* and *event* entities only; items for papers, scholars, institutions and event series rely on complementary community curation and are therefore not uniformly complete. Second, the reconciliation pipeline depends on the availability and quality of external endpoints (DBLP, K10plus, geograpy3); degradation or schema changes in these sources propagate into the pipeline. Third, roughly 20 % of CEUR-WS volumes are not yet indexed by DBLP, which limits editor and author disambiguation for that subset. Fourth, the current Single Point of Truth remains the legacy HTML/PDF representation; the metadata-first workflow outlined in Section 8 has not yet replaced it in production. Finally, generalisation beyond CEUR-WS to other publishers has not been empirically evaluated and is left for future work.

9.2 Supplemental Material Statement

Source code for the prototypes and semantification tools is available via the repositories of the github ceurws organization[21]. Extended supplemental material — including the schema diagrams (Figures 1 and 2) in their editable Graphviz form, the SPARQL queries derived from the Semantic Publishing Challenge, and background research — is provided by the CR research wiki [15]. The LaTeX sources, bibliography, build pipeline and the `AGENTS.md` governance policy for AI-assisted editing of this paper are available at [21]. The cited software artefacts are released under Apache-2.0: `ceur-spt` [18], `pyCEURmake` [17], `snapquery` [19], and `wdgrid` [16].

9.3 Declaration of use of Generative AI

The authors used AI-based coding assistants during the engineering and preparation of this work. Use of such assistants is governed by an AGENTS.md policy file maintained in this paper's repository and, equivalently, in the source repositories of the software artefacts ceur-spt [18] and pyCEURmake [17]. The policy mandates:

[21] https://github.com/ceurws

– a plan-and-confirm protocol: the assistant must state its analysis, plan, and scope, and receive explicit human confirmation before any read, write, or execution;
– a strict scope for the manuscript itself: AI assistance is limited to spelling/grammar corrections and formatting support (tables, figures, build automation). AI was not used to generate, rewrite or contribute original scientific content, arguments, or prose;

All scientific claims, interpretations and conclusions are the sole responsibility of the authors.

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